TEXORA · LECTURE SHEET

Advances in Knitting Technology

From mechanical pattern chains to computer-controlled needles and whole garments off the machine: electronic selection, CAD pattern preparation, seamless and whole-garment knitting, technical and 3D knitting, and the connected, monitored knitting mill.

FM-4201 · ADVANCED KNITTING AND NON-WOVEN · TOPIC 1 · FABRIC MANUFACTURING SPECIALIZATION

1. The direction of travel

The basics of TE-3101 — needles, cams, sinkers, feeders — have not changed in principle. What has changed is **who or what makes each decision**: pattern decisions moved from pattern wheels and chains to electronic needle selection under computer control; shaping decisions moved from the cutting table to the knitting programme; and quality decisions are moving from post-production inspection to in-process monitoring. This sheet follows those three movements, then the product families they created.

2. Electronic selection and computer control

- **Electronic needle selection:** on modern circular and flat machines each needle (or jack) is selected by electromagnetic or piezoelectric actuators course by course, so jacquard patterns of effectively unlimited repeat, intarsia fields and structure changes are programmed, not chained. Pattern preparation is done in CAD: design, colourway, structure simulation and machine file from one workstation.
- **Individual yarn-feed control:** positive feeders with electronic metering hold stitch length (the master variable of TE-3205) to target by measuring yarn input, not by trusting cam settings — fabric weight consistency is the direct beneficiary.
- **Machine productivity:** feeder counts and speeds on large-diameter circular machines have climbed together: a 96-feeder machine at 25 rpm forms 96 × 25 = **2400 courses per minute** — the arithmetic that, with course density, sets output (FM-3207 calculations), and that makes electronic reliability, not mechanism, the limit on production.
- **Fine and coarse extremes:** needle counts follow gauge and diameter — a 30-inch, 28-gauge cylinder carries π × 30 × 28 ≈ **2639 needles**, each one individually selectable on a jacquard machine;

at the other extreme, coarse-gauge flat machines knit yarns that once only hand knitting could handle.

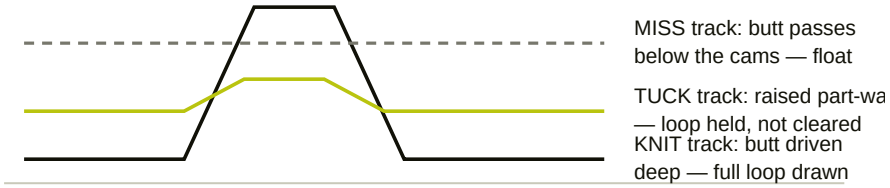


Figure 1 — One feeder, three decisions: the depth of the cam track at the stitch point decides whether the needle knits, tucks or misses. Mechanical machines fix these paths in steel for a whole machine; electronic selection gives every needle its own decision at every feeder — the advance this sheet is about.

3. Seamless and whole-garment knitting

Circular seamless (small-diameter machines with multiple feeds and electronic selection) knits body-size tubes already carrying their pattern, elastics and zones — underwear, sportswear, hosiery-type products with no side seams and minimal make-up. **Whole-garment flat knitting** (the technology commercialised as WHOLEGARMENT and knit-and-wear) knits the complete garment — body and sleeves as connected tubes shaped by transfer and partial knitting — so the product leaves the machine needing only minimal linking/finishing. The advances that made it possible: fast reliable stitch transfer, holding-down sinkers that let needles start and stop within a course, compact carriages with independent systems, and programming systems that simulate the garment before a cone is mounted.

WORKED EXAMPLE — THE WASTE ACCOUNT THAT DRIVES WHOLE-GARMENT ADOPTION (ILLUSTRATIVE)

Take a garment containing 0.85 kg of yarn as worn.

Cut-and-sew from knitted panels: panels are knitted at 1.00 kg and cut to shape; cutting waste ≈ 15% → yarn input **1.00 kg** per garment, plus sewing labour for every seam.

Whole-garment knitting: the shaped garment is knitted almost to size; trim and linking allowance ≈ 2% → input 0.85 ÷ 0.98 ≈ **0.87 kg**.

Saving ≈ 0.13 kg of yarn per garment (13% of input) and most cutting and seaming operations — the twin economic and sustainability argument for the technology, before any quality claim is made.

4. Technical and 3D knitting

The same control advances turned knitting into a technical-textile former: **spacer knitting** (FM-4103 Topic 2) on double-bed machines; **inlay and weft-insertion** structures that lay straight load-bearing yarns into a knitted net for composites and geotextiles; **multiaxial warp knitting**, inserting yarn sheets at several angles bound by a chain stitch — the reinforcement fabrics of the composites industry; **shaped preforms** — knitted tubes that branch, bend and vary diameter for medical (stents, vascular grafts), automotive and composite parts; and **knitted e-textiles** (FM-4103 Topic 4), where selection places conductive yarns as circuits and sensors.

5. The connected mill: monitoring and Industry 4.0

Modern knitting plants network their machines: production, efficiency, stop causes and yarn consumption stream to central systems; fabric inspection moves toward camera systems on or off the machine that grade defects in real time (the defect vocabulary of FM-4102 Topic 3, automated); predictive maintenance watches vibration, temperature and stoppage patterns; and digital product data (the design file, the machine file, the inspection record) increasingly follow the fabric roll as its passport. For the technologist, the skill shift is the point: pattern programming, data interpretation and process engineering gain weight against purely mechanical craft — without replacing it, since the machine still stops for the same broken needle it always did.

6. Sustainability as a technology driver

Knitting's structural advantage — it forms shaped fabric directly from yarn, with no weaving preparation and little cutting waste — is being developed deliberately: whole-garment production (Section 3), dope-dyed and recycled yarns knitted without piece dyeing, on-demand/short-run production enabled by fast electronic changeover, and design-for-disassembly in technical knits. The counterweights are equally structural: electronic machines are capital-intensive, programming skill is scarce, and fine-gauge high-speed knitting is unforgiving of poor yarn — advances arrive as a system (machine + yarn + software + skill), never as a machine alone.

Glossary

Electronic needle selection — per-needle, per-course patterning control by actuators under computer command. **Whole-garment knitting** — complete garment knitted in one piece on a flat machine. **Seamless (circular)** — body-size tube products knitted without side seams. **Multiaxial warp knitting** — inserted yarn layers at several angles bound by chain stitches. **Positive feed** — metered yarn delivery controlling stitch length directly.

Key points

• The advances are about decisions: pattern → electronics and CAD; shaping → the knitting programme; quality → in-process monitoring.
• Scale check: 96 feeders × 25 rpm = 2400 courses/min; a 30″ 28G cylinder carries ≈2639 needles.
• Whole-garment economics are a waste account: ≈13% yarn saving in the illustrative example, plus removed cutting/seaming.
• Technical knitting (spacer, inlay, multiaxial, shaped preforms, e-textiles) is the growth frontier — the machine as a former of structures, not just cloth.

Practice

1. A circular jacquard machine has a 34-inch cylinder of 24 gauge. Estimate its needle count.
2. State two distinct capabilities that electronic needle selection gives a designer that a pattern-wheel machine cannot match.
3. Why does positive yarn feed improve fabric weight consistency more than cam adjustment alone? Answer through stitch length.
4. Give one product family each for circular seamless and for whole-garment flat knitting, with the structure reason the route suits it.

Answer key

1. Needles = π × 34 × 24 ≈ **2564 needles**.
2. (i) **Unlimited pattern repeat** — every course can differ, so true jacquard pictures/lettering and garment-size pattern panels are possible rather than small repeating motifs. (ii) **Course-by-course structure change** — knit/tuck/miss and yarn changes placed exactly, enabling intarsia fields, inlaid zones and shaped/seamless programmes in one fabric.
3. Fabric weight is set by stitch length (GSM identity, TE-3205): GSM follows l exactly. A cam only *sets* a nominal stitch length — actual yarn input drifts with tension, yarn friction and package variation. A positive feeder *measures and meters* the yarn delivered per course, closing the loop on l itself, so the variable that determines weight is the variable being controlled.
4. Circular seamless: **sports base-layer tops** — body tube with knitted-in ventilation zones and elastics, no side seams to chafe. Whole-garment flat: **fashion sweaters** — sleeves and body knitted connected and shaped (Topic 2), removing panel cutting and most seaming while allowing fashioned detail flat machines alone can shape.